

# **BIP-110**

Proposed changes and dates

# Overview

# Metadata

| Field         | Value                           |
|---------------|---------------------------------|
| BIP           | 110                             |
| Title         | Reduced Data Temporary Softfork |
| Layer         | Consensus (soft fork)           |
| Type / status | Specification / Draft           |
| Author        | Dathon Ohm                      |
| Assigned      | 2025-12-03                      |
| Snapshot date | 2026-05-26                      |

# Summary

- BIP-110 proposes temporary consensus checks against several large-data transaction patterns
- The checks apply during an active window of 52,416 blocks, roughly one year
- Inputs spending UTXOs created before activation are exempt
- After expiry, the added restrictions stop applying
- This deck describes the proposal, not whether it should activate

# Proposed checks

| Object                               | Restriction while active                                                  |
|--------------------------------------|---------------------------------------------------------------------------|
| scriptPubKey                         | New output scripts over 34 bytes invalid, except OP_RETURN up to 83 bytes |
| OP_PUSHDATA\*                        | Payloads over 256 bytes invalid, except BIP-16 redeemScript push          |
| Script argument witness items        | Items over 256 bytes invalid                                              |
| Undefined witness / Tapleaf versions | Spending them invalid                                                     |
| Taproot annex                        | Witness stacks with an annex invalid                                      |
| Taproot control block                | Blocks over 257 bytes invalid                                             |
| Tapscript opcodes                    | OP_SUCCESS\* invalid; executed OP_IF / OP_NOTIF invalid                   |

# Technical Context

# Consensus rule, not relay policy

- Relay policy controls mempool acceptance and transaction forwarding
- Consensus controls whether a block is valid
- BIP-110 is a consensus proposal
- If active, enforcing nodes reject blocks containing violating transactions
- Non-upgraded nodes still validate those blocks using pre-BIP-110 rules

# Outputs and scriptPubKeys

- A transaction output contains an amount and a scriptPubKey
- The scriptPubKey locks the output to a spending condition
- Unspent outputs live in the UTXO set
- Common output forms already fit within 34 bytes: P2PKH 25, P2SH 23, P2WPKH 22, P2WSH 34, P2TR 34
- OP\_RETURN is provably unspendable and gets a separate 83-byte allowance

## Pushes, scripts, and witness

- Script push opcodes place byte strings on the execution stack
- OP\_PUSHDATA1, OP\_PUSHDATA2, and OP\_PUSHDATA4 support explicitly sized payloads
- SegWit witness data carries signatures, scripts, and script arguments outside the legacy txid
- BIP-110 caps large contiguous pushes and script argument witness items at 256 bytes
- It does not simply cap all scripts at 256 bytes

# Taproot objects

- A Taproot output can be spent by key path or script path
- Script-path spending reveals one Tapleaf script and a control block
- The control block proves that the revealed leaf is in the committed Taptree
- Control block size is  $33 + 32 * \text{Merkle-depth}$
- The annex is a currently undefined optional Taproot witness element

# Tapscript hooks

- BIP-342 defines Tapscript for Taproot script-path spends
- `OP_SUCCESSx` opcodes are reserved upgrade hooks that currently make validation succeed
- `OP_IF` and `OP_NOTIF` provide conditional branching
- BIP-110 temporarily restricts both surfaces
- This is one reason the BIP is temporary: the rules constrain future upgrade mechanisms

# **Proposed Rules**

## Rule 1: output scripts

- New output scriptPubKeys over 34 bytes are invalid
- Exception: OP\_RETURN outputs may be up to 83 bytes
- Standard payment output forms fit within the limit
- Large spend conditions can still be committed by hash or Taproot output key
- The direct target is large spendable output scripts

## Rule 2: large data chunks

- `OP_PUSHDATA\*` payloads over 256 bytes are invalid
- Script argument witness items over 256 bytes are invalid
- The BIP-16 redeemScript push in P2SH scriptSigs is exempt
- Revealed witness scripts and Tapleaf scripts are not script argument witness items
- Their executed pushes are still subject to the push-size rule

## **Rule 3: undefined versions**

- Spending undefined witness versions is invalid
- Spending undefined Tapleaf versions is invalid
- Defined versions remain valid: SegWit v0, Taproot, and P2A as named by the BIP
- Creating outputs with undefined witness versions remains valid
- Technical effect: this upgrade space cannot be spent during the active period

## **Rules 4 and 5: annex and control blocks**

- Witness stacks with a Taproot annex are invalid
- Taproot control blocks larger than 257 bytes are invalid
- 257 bytes means maximum Merkle depth 7
- A balanced depth-7 Taptree can contain up to 128 leaves
- Deep or deliberately unbalanced Taptrees may be affected

## Rules 6 and 7: Tapscript opcodes

- Tapscripts containing `OP_SUCCESS\*` anywhere are invalid
- This applies even if the opcode would not otherwise execute
- Tapscripts executing `OP_IF` or `OP_NOTIF` are invalid
- The condition result does not matter
- Some Miniscript-generated Tapleaves may need adjustment

## Grandfathering and expiry

- Inputs spending UTXOs created before activation are exempt from all new checks
- New outputs created after activation are checked at creation and, where applicable, when spent
- This matters because Taproot trees are hidden at output creation time
- After expiry, UTXOs of all heights are unrestricted again
- The proposal is height-sensitive, not only transaction-format-sensitive

# What stays unchanged

- Monetary supply rules
- Proof-of-work and block header format
- Block weight limit
- Ordinary P2PKH, P2SH, P2WPKH, P2WSH, and P2TR outputs
- Taproot key-path spends
- Commitment-by-hash patterns

# Deployment

# Parameters

| Parameter             | Value                      |
|-----------------------|----------------------------|
| Deployment name       | reduced_data               |
| Version bit           | 4                          |
| Start time            | 2025-12-01 00:00:00 UTC    |
| Timeout               | NO_TIMEOUT                 |
| Max activation height | 965,664                    |
| Active duration       | 52,416 blocks              |
| Threshold             | 1,109 / 2,016 blocks (55%) |

# State machine

1. **DEFINED**: before start time
2. **STARTED**: miners may signal bit 4
3. **LOCKED\_IN**: threshold reached, or forced no later than height 963,648
4. **ACTIVE**: one retarget period after lock-in; new rules enforced
5. **EXPIRED**: activation height plus 52,416 blocks; new rules stop applying

## Differences from BIP-9

| <b>Standard BIP-9</b>  | <b>BIP-110</b>                                |
|------------------------|-----------------------------------------------|
| 95% threshold          | 55% threshold                                 |
| Timeout can fail       | NO_TIMEOUT; no ordinary FAILED state          |
| No mandatory signaling | Mandatory signaling before maximum activation |
| ACTIVE is terminal     | ACTIVE later becomes EXPIRED                  |
| Permanent once active  | Temporary: 52,416 active blocks               |

# Important heights and dates

| When                    | Height              | Meaning                    |
|-------------------------|---------------------|----------------------------|
| 2025-12-01 00:00 UTC    | n/a                 | Start time                 |
| 2025-12-03              | n/a                 | BIP assigned               |
| Aug 2026                | 961,632-963,647     | Mandatory signaling window |
| Aug 2026                | 963,648             | Latest lock-in boundary    |
| Sep 2026                | 965,664             | Latest activation height   |
| 1 year after activation | activation + 52,416 | Expiry                     |

## Activation paths

- Early path: any retarget period reaches 1,109 signaling blocks
- Mandatory path: blocks 961,632 through 963,647 must signal bit 4
- Lock-in to activation is one retarget period, about two weeks
- If activation is at 965,664, expiry is at 1,018,080
- Calendar dates are estimates; heights are the exact consensus boundaries

**Effects**

## Affected patterns

- Large data in spendable output scripts
- Large contiguous script pushes
- Large script argument witness items
- Undefined witness and Tapleaf version spends
- Taproot annex usage
- Very deep Taproot script trees
- Tapscripts using `OP_SUCCESS*`, `executed OP_IF`, or `executed OP_NOTIF`

## Wallet and contract notes

- Wallets creating post-activation Taproot scripts must avoid banned constructs
- Miniscript compilers may need temporary policy changes
- Protocols using large Taptrees may need shallower constructions
- Pre-signed Taproot transactions need attention if they must confirm and spend during the active period
- The BIP describes narrow possible fund-freezing or fund-loss scenarios

# Sources

## Primary sources

- BIP-110: Reduced Data Temporary Softfork
- BIP-9: Version bits with timeout and delay
- BIP-16: Pay to Script Hash
- BIP-141: Segregated Witness consensus layer
- BIP-341: Taproot
- BIP-342: Tapscript

Sources

# Q&A